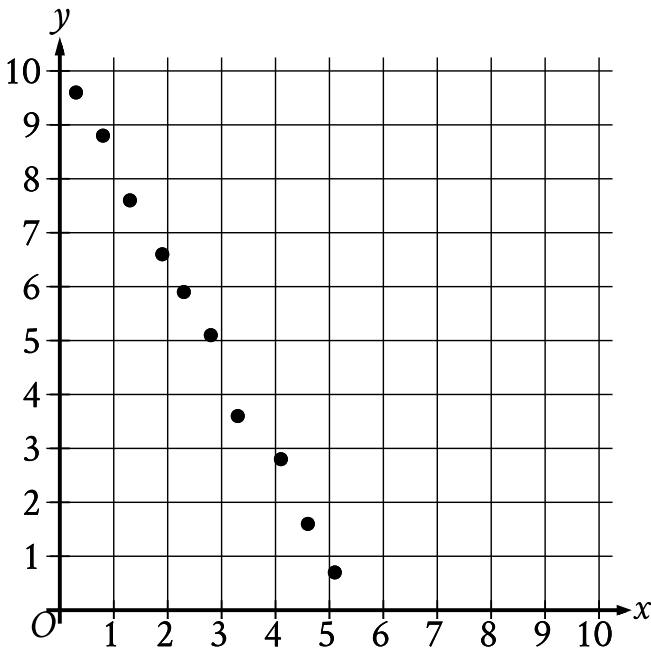


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Easy

Question



Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

- Answer
- A. $y = -1.9x - 10.1$
 - B. $y = -1.9x + 10.1$
 - C. $y = 1.9x - 10.1$
 - D. $y = 1.9x + 10.1$

Correct Answer: B

Rationale

Choice B is correct. The equation representing a linear model can be written in the form $y = a + bx$, or $y = bx + a$, where b is the slope of the graph of the model and $(0, a)$ is the y-intercept of the graph of the model. The scatterplot shows that as the x-values of the data points increase, the y-values of the data points decrease, which means the graph of an appropriate linear model has a negative slope. Therefore, $b < 0$. The scatterplot also shows that the data points are close to the y-axis at a positive value of y . Therefore, the y-intercept of the graph of an appropriate linear model has a positive y-coordinate, which means $a > 0$. Of the given choices, only choice B, $y = -1.9x + 10.1$, has a negative value for b , the slope, and a positive value for a , the y-coordinate of the y-intercept.

Choice A is incorrect. The graph of this model has a y-intercept with a negative y-coordinate, not a positive y-coordinate.

Choice C is incorrect. The graph of this model has a positive slope, not a negative slope, and a y-intercept with a negative y-coordinate, not a positive y-coordinate.

Choice D is incorrect. The graph of this model has a positive slope, not a negative slope.